Low-Resource Machine Translation on Low Resources

Gabriel Selzer
UW–Madison
Madison, WI, USA
gjselzer@wisc.edu

Luke Neuendorf
UW–Madison
Madison, WI, USA
lneuendorf@wisc.edu

Zach Potter
UW–Madison
Madison, WI, USA
zmpotter@wisc.edu

Abstract

Much like the rest of natural language processing research, low-resource machine translation (MT) research has trended towards larger, more powerful models to achieve higher benchmarks on low-resource language translation tasks. This poses concerns for accessibility, since larger models require proportionally powerful hardware for training, testing, and execution - this compute demand cannot yet be feasibly replicated across many low-resource languages. We therefore believe there is still value in smaller, easily re-trainable models for low-resource MT. We evaluate five different low-resource MT techniques not yet tested for low-resource language translation on an NVIDIA 2080 Ti GPU with a maximum of 3 hours of training time, simulating a more accessible training environment. We show the mBART principle adds significant gains in translation quality across both tested translation tasks. RASP, Joint Dropout, and Part-of-Speech tags enabled significant gains on the translation of linguistically similar language pairs but failed to reproduce those gains on diverse language pairs. Data Diversification negatively impacted translation performance for both tested translation tasks.

1 Introduction

Machine translation (MT), the procedure of automating natural language translation, can enable speakers of one language to freely access information and entertainment written in languages they cannot understand. The potential of a machine translator whose accuracy rivals that of a human translator is particularly appealing for speakers of "low-resource" languages, who could freely gain access to writings in "high-resource" languages spoken by many more people around the world.

Training machine translators requires "bitexts", large corpora of parallel sentences in the source and target languages, but of the more than 7000 languages (Anderson, 2010) spoken around the world, very few are well represented in available bitexts, and the largest bitexts usually belong to high-resource language pairs (Haddow et al., 2022). Until larger bitexts become available for low-resource languages, advancements in low-resource MT generally rely on specialized models, and the benefits of large language models including BERT (Devlin et al., 2019), GPT-3 (Brown et al., 2020), and their derivatives are unavailable to speakers of low-resource languages. To avoid leaving speakers of low-resource languages behind, the translation of low-resource languages has become an active area of research, and low-resource translation shared tasks have been common at prominent MT conferences (Weller-di Marco and Fraser, 2022) (Anastasopoulos et al., 2022)

Both within and outside of the shared tasks, however, the majority of low-resource MT advancements come from utilizing larger models. Made feasible by more powerful, newer, and more expensive hardware, including Graphics Processing Units (GPU)s and Tensor Processing Units (TPU)s, such models use hundreds of millions to hundreds of billions of parameters to achieve new benchmarks in low-resource translation. For example, in the WMT 2022 low-resource MT Shared Task, the strongest system trained a Transformer with 35 encoder layers and the second place system trained mBART from scratch (Weller-di Marco and Fraser, 2022). While these achievements in translation quality are impressive, they can hardly be replicated at scale across a wide selection of low-resource languages, especially for languages whose speakers are located in low-income regions of the world. We believe that there is still inherent value in advancing low-resource MT quality without increasing the model's size or increasing its training time - this forwards the goals of accessibility, rapid iteration of new techniques, and execution on cheaper devices. Put simply, the barrier of limited data is compounded by the barrier of limited technology in creating accessible, widespread, high-quality MT for the world's low-resource languages.

We believe that the key to unlocking accessible, low-resource MT is to enable training within the limits imposed by consumer-grade hardware. To enable effective training on laptops and desktops owned by university students and researchers would enable far more people to create translators and positively impact communities of low-resource language speakers. Because low-resource MT is such a popular area of research with many existing, proven solutions, we evaluate a set of recent innovations in low-resource MT based on their performance within a framework devised for accessibility. Some of these innovations were evaluated by training on down-sampled data from a high-resource language bitext and utilize additional resources only available for high-resource languages. For these techniques, we wish to determine whether they still succeed when those additional resources are less robust. Other innovations were tested on models beyond the limits of what could be trained on consumer-grade hardware. In these cases, we wish to determine whether a technique would still succeed on a smaller model, trained with fewer passes over the data. Both cases highlight the accessibility of training a MT model from scratch, which could be performed on a consumer laptop in a few hours by anyone.

As a part of this evaluation, we also consider translation between languages of varying linguistic similarity to determine the robustness of these techniques regardless of language. Finally, we evaluate the set of methods shown to be effective on our smaller model in a final composition model to determine whether the independent effects can compound for greater benefit.

2 Related Works

Low-resource MT is a very active area of research, however very little has focused on smaller models. The largest hurdle towards low-resource MT lies in assembling a parallel language corpus, as by definition, low-resource languages offer little choice between small corpora of parallel data (Haddow et al., 2022). To that end, two major techniques are used to provide additional information, with documented improvements in low-resource translation. Back-translation enables the synthesis of additional training data by first training an intermediate model to translate a high-resource language into a low-resource language, producing a new bitext through the translation of additional monolingual corpora (Chen et al., 2020). The back-translation process has been iterated on since its release, and many models today use Data Diversification, a technique implementing both back-translation and forward-translation to generate an even larger synthetic corpus (Nguyen et al., 2019). The second technique utilized is to incorporate bilingual dictionaries into the training phase or the loss calculation, used to reinforce word relations. (Lin et al., 2020) found that replacing words in source sentences with their translations with a 30% probability resulted in impressive performance gains on both low-resource and high-resource languages.

An important choice in building MT models lies in architecture selection and most modern models rely on pre-trained multilingual machine translators (Haddow et al., 2022). mBART, an extension of the BART model trained on 25 languages from the Common Crawl, has become a popular choice in recent years, as the architecture provides benefits to low-resource and multilingual settings. mBART differentiates itself by masking 35% of the input tokens on a provided sentence during training and attempting to recover the original sentence, minimizing loss across all languages simultaneously. The language itself is identified using a tag token, and in fine-tuning parallel corpora are used to train the model to translate (Liu et al., 2020). The dominance of mBART and similar massive multilingual models seems to stem from their ability to gather a shared representation of meaning across many languages - this translates well into low-resource target languages (Haddow et al., 2022). The classic alternative to mBART is the Transformer architecture (Vaswani et al., 2017), although the majority of implementations use a significantly larger model than described in the original paper. For example, the HW-TSC system, which scored very well in the WMT low-resource shared tasks in 2022 (Weller-di Marco and Fraser, 2022), is based on a version of fairseq's Transformer-Big architecture, but with 35 transformer encoder layers instead of the standard 6 layers (Li et al., 2022).

In low-resource situations, where morphological and syntactic information may not be prevalent enough for a model to learn, external resources can be useful in providing extra information to the model. One large pitfall involves subword tokenization, a procedure with near-universal adoption that builds subword units using algorithms such as Byte-Pair Encoding (BPE). (Ding et al., 2019) showed the performance of algorithms like BPE vary drastically in low-resource settings, and that there is no single number of merge operations that works equally well across different languages and bitexts. There are several interesting alternative methods of incorporating linguistic information into the model; one alternative, useful when fine-tuning on a low-resource source with a different word order than the parent model source, involves reordering the parent source language sentences to conform to the child source's word order. This alternative was shown to have statistical significance in many different experiments (Murthy et al., 2019). Another alternative, thoroughly tested by (Sánchez-Cartagena et al., 2020), involves interleaving either part-of-speech tags or morpho-syntactic description tags into the source and/or target sentences; this lexical information was shown to improve translations, especially when the tags augmented the morphologically poorer information.

There are many efforts directed towards increasing the accessibility of larger models. Many businesses offer cloud-based model training at cost for researchers and other businesses alike. These systems provide server-grade hardware for a fraction of the price and enable portability, but they also require an internet connection and continued funding. Another interesting concept is that of model distillation, wherein a larger model already trained to produce desired results is used to train a smaller model to reproduce those accurate results (Hinton et al., 2015). While this technique can be used to enable a model to be run on cheaper, weaker devices, and can retain accuracy as shown by (Mohammadshahi et al., 2022), it requires the larger model be trained in the first place - it does not remove the powerful machine from the training loop.

3 Methods

Our goal is to enable accessible low-resource MT, and to that end we set computational baselines that we believe are accessible to far more researchers and developers. Specifically, we limit our experiments to training on an NVIDIA 2080 Ti GPU and limit training to a maximum of 3 hours. Seeking improvements within this framework, we examine five different techniques. Each technique was published within the last four years, and each showed the potential to transfer to our particular framework. With the exception of Joint Dropout, which was the focus of our re-implementation in our project proposal, none of these systems have been studied before in our framework, either being tested with more data, on a model with more parameters, or both. For each technique, we transfer the innovative principle to our framework. In all cases, this required re-implementation for execution on top of a shared Transformer architecture. Each implementation is described below.

3.1 Joint Dropout

Joint Dropout (JD) is a data-centric technique aimed at enhancing the generalization and robustness of Neural Machine Translation (NMT) models without altering their architecture (Araabi et al., 2023). It works by first generating word alignments across the bitext. Then, aligned phrases in source and target sentences are randomly dropped and replaced with joint variables. Phrases can be composed of several words. For example, in the English-to-German translation ⟨*We ate apples by the ocean., Wir haben Äpfel am Meer gegessen.*⟩, the phrase "ate" aligns with "gegessen" and the phrase "We" aligns with "Wir." Thus, JD can be performed by adding variables in the corresponding positions as ⟨*X1 X2 apples by the ocean., Y1 haben Äpfel am Meer Y2.*⟩. The goal of this approach is to aid the model in interpreting sentences without bias toward specific words or expressions.

The authors of (Araabi et al., 2023) tested JD using the IWSLT 2014 shared task (Cettolo et al., 2014) German-English translation dataset and two transformer models (Transformer-base and Transformer-opt) from the fairseq library (Ott et al., 2019). Furthermore, they introduced the Joint Dropout Rate (JDR), the ratio of dropped to total tokens in both source and target sentences, and determined the optimal JDR to be 0.3. They simulated a low-resource setting by randomly sampling 10,000 sentence translation pairs from the training corpus. Their implementation with the Transformer-opt model yielded an 18.0 sacreBLEU score without JD versus a 19.9 sacreBLEU score with JD. We re-implemented JD in our project proposal for comparative analysis and to evaluate its performance on various translation pairs.

We added JD to our framework using the Eflomal (Efficient Low-Memory Aligner) library (Östling and Tiedemann, 2016) to generate word alignments between sentence pairs and a fork of the phrase extraction function in the Eliezer library (Tan, 2014) to generate phrase translation tables from the alignments.

3.2 Data Diversification

Data Diversification is a semi-supervised data augmentation technique that builds upon the idea of back-translation (Nguyen et al., 2019). Like back-translation, Data Diversification trains intermediate models: a forward translation model and a backward translation model. Unlike back-translation, Data Diversification uses the original training corpus to synthesize new data. This is done by passing the source training data through the forward-translation model to generate synthetic target text and the target training data through the back-translation model to generate synthetic source text. These synthetic texts are paired with their original source and target training texts, respectively. This process can be repeated several times and is stated more formally below.

Algorithm 1 Data Diversification

Parameters:
- $D = (S, T)$ - Training bitext
- k - Diversification factor
- N - Number of rounds

procedure DIVERSIFY($D = (S, T), k, N$)
  $D_0 \leftarrow D$
  for $r \in 1, \ldots, N$ **do**
    $D_r = (S_r, T_r) \leftarrow D_{r-1}$
    for $i \in 1, \ldots, k$ **do**
      $M^i_{S \to T,r} \leftarrow Train(S_{r-1}, T_{r-1})$
      $M^i_{T \to S,r} \leftarrow Train(T_{r-1}, S_{r-1})$
      $D_r \leftarrow D_r \cup (S, M^i_{S \to T,r}(S))$
      $D_r \leftarrow D_r \cup (M^i_{T \to S,r}(T), T)$
  $\hat{M}_{S \to T} \leftarrow Train(D_N)$
  return $\hat{M}_{S \to T}$

(Nguyen et al., 2019) conducted three experiments: two in high-resource settings and one in a low-resource setting. The first high-resource test was conducted on the WMT'14 English-German (En-De) and English-French (En-Fr) (Bojar et al., 2014), whose training sets had 27M and 136M sentence pairs, respectively. The second high-resource task was conducted using the IWSLT'14 German-English (De-En) dataset (Cettolo et al., 2014) and the IWSLT'13 French-English (Fr-En) dataset (Cettolo et al., 2013), with 160K and 200K training sentence pairs, respectively. The third experiment, tested on low resources, was conducted on the FLORES English-Nepali (En-Ne) and English-Sinhala (En-Si) datasets (Guzmán et al., 2019), each containing about 500K and 400K sentences pairs, respectively.

The two high-resource experiments saw BLEU score increases of at least 1.4 for the En-De translations and 1.0 BLEU for the Fr-En translations. The low-resource experiment saw sacreBLEU scores increase by 1.3 for Ne-En and by 1.5 for Si-En. In our experiments, we will evaluate whether Data Diversification can improve translation performance with far fewer training examples.

3.3 mBART

Fundamentally, BART (Lewis et al., 2019) corrupts text using a noising function and then learns a model that can reconstruct the original text. mBART (Liu et al., 2020) expands upon BART by de-noising corpora containing multiple languages.

We applied the principles of mBART's loss function, shown in Equation 1, to our pre-processing pipeline in order to implement this pre-training objective. In particular, given a multilingual corpus D with sentences X, we add noise to X using a noising function $g(X)$ and pre-train our model to maximize $P(X|g(X))$. Our noising function mirrors that of (Liu et al., 2020), in that we mask 35% of the words in each sentence according to a Poisson distribution ($\lambda = 3.5$). For example, given sentences ⟨*We ate apples by the ocean., Wir haben Äpfel am Meer gegessen.*⟩, masking may yield the sentences ⟨*_ _ apples by the ocean., _ _ Äpfel am Meer gegessen.*⟩.

$$\mathcal{L}_\theta = \sum_{X \in D} log(P(X|g(X); \theta)) \quad (1)$$

(Liu et al., 2020) experimented with mBART on 24 En-X bitexts representing a wide range of available sentence pairs (as low as 10k for English ↔ Gujarati to as high as 4.50M for English ↔ Latvian) on a standard sequence-to-sequence Transformer architecture (680M parameters). They achieved gains with mBART on all of their low-resource and medium-resource language pairs compared to their respective baselines, with BLEU increases of as much as 12+ on some of their low-resource pairs. However, in contrast to (Liu et al., 2020), who trained their mBART model over the course of 2.5 weeks using 256 32GB NVIDIA V100 GPUs, we evaluate whether the principles of mBART can continue to show gains for low-resource language pairs when trained on a consumer machine in just a matter of hours.

3.4 mRASP

Multilingual Random-Aligned Substitution Pre-training (mRASP) adds a pre-training objective to minimize the differences between representations of similar phrases in different languages (Lin et al., 2020). mRASP's fundamental contribution, Random Aligned Substitution (RAS), is shown in Equation 2 and requires bitexts (S, T) containing sentences (S^i, T^i). For each S^i, RAS applies a dictionary function $C_S(S^i)$ to randomly replace some words of S^i with their translations in another language U.

$$\mathcal{L}_\theta = \sum_{(S,T)} \sum_{(S^i,T^i)\in(S,T)} [-log(P(T^i|C_S(S^i);\theta))] \quad (2)$$

As an example, again consider the English-German sentence pair ⟨*We ate apples by the ocean., Wir haben Äpfel am Meer gegessen.*⟩. During pre-training, we might replace "apples" with "Äpfel" and "the" with "das", yielding the pre-training sentence pair ⟨*We ate Äpfel by das ocean., Wir haben Äpfel am Meer gegessen.*⟩. After pre-training, we train using the original sentence pair.

(Lin et al., 2020) applied mRASP over 197M parallel sentences across 32 En-X bitexts and tested on the Transformer-Large architecture (∼100M parameters). Each En-X bitext corresponded to a dictionary within the MUSE project (Lample et al., 2018). While (Lin et al., 2020) showed impressive gains on their model, showing an average increase of 22.5 BLEU on low-resource language pairs, we will evaluate whether RAS can benefit an even smaller model, trained on a single bitext with fewer parallel sentences and using dictionaries with fewer entries.

To implement mRASP, we require a bilingual dictionary for each language pair tested. Like (Lin et al., 2020), we use MUSE to access a German-English dictionary, containing over 78K English words and their German translations. We use the bilingual dictionary provided in the Nepdict project (Nirooj et al., 2020) as our Nepali-English dictionary, which contains 975 pairs of English words along with one or more translations in Nepali.

3.5 POS Tags

Linguistic tag Interleaving seeks to improve translation quality by appending a tag to each token in a sentence. While tag interleaving has been studied in MT for decades, (Sánchez-Cartagena et al., 2020) were the first to extensively compare different tag types on the Transformer architecture, including part-of-speech (POS) and morpho-syntactic description (MSD) tags. To continue using our example sentence pair ⟨*We ate apples by the ocean., Wir haben Äpfel am Meer gegessen.*⟩, tagging the source sentence with POS tags yields the sentence ⟨*We <PRON> ate <VERB> apples <NOUN> by <ADP> the <DET> ocean <NOUN> . <PUNCT>, Wir haben Äpfel am Meer gegessen.*⟩.

(Sánchez-Cartagena et al., 2020) applied interleaved tags to the base Transformer architecture and showed statistically significant increases in translation quality, but tested on bitexts with a minimum of 50k randomly selected parallel sentences, and only studied languages well supported by the Stanza NLP package (Qi et al., 2020). We will evaluate whether interleaving tags within the Transformer model can create similar benefits for even less training data, and on languages without support from an existing Tagger application. For accessibility, we restrict our experiments to source-language POS tags and do not add tags to the target language.

To include interleaved POS tags in our datasets, we require a POS tagger for each source language tested. We use the Stanza POS Tagger to tag German source sentences (Qi et al., 2020). To tag Nepali source sentences, we train our own POS Tagger using `scikit-learn` (Pedregosa et al., 2011) due to the lack of an open, accessible POS Tagger for Nepali. We train and test this POS Tagger on existing tagged Nepali documents within the FLORES dataset, containing 47k Nepali words each along with their part of speech, and obtain an accuracy of 92%.

4 Experiments

4.1 Data

To evaluate the capabilities across language pairs with varying linguistic similarity, we trained each model on two different datasets. We first gather German-English (De ↔ En) translation data from the IWSLT 2014 shared task (Cettolo et al., 2014), representing linguistically similar languages, and we then gather Nepali-English (Ne ↔ En) data from the FLORES evaluation datasets (Guzmán et al., 2019). To realize the computational constraints described in Section 3, we artificially sample each dataset to 10 thousand and 20 thousand samples, respectively. The testing and validation datasets used to test each translation model were gathered from their respective sources and used unaltered.

4.2 Model

We utilize the same transformer configuration for all experiments. We chose the Base Transformer model from the fairseq library (Ott et al., 2019), chosen for its accessibility, simplicity, and performance. Notably, we utilize the `transformer_iwslt_de_en` architecture, which is sufficiently small to be trainable within our computational scenario. Following the lead of (Araabi et al., 2023), we make the adjustments shown in Table 1, reducing the model to between 36 and 40 million parameters, depending on the vocabulary size of the dataset.

Hyperparameter	Value
`-encoder-layers`	5
`-decoder-layers`	5
`-encoder-attention-heads`	2
`-decoder-attention-heads`	2
`-encoder-ffn-embed-dim`	1024
`-decoder-ffn-embed-dim`	1024
`-decoder-layerdrop`	0.2
`-activation-dropout`	0.3
`-label-smoothing`	0.5

Table 1: Hyperparameter alterations necessary to alter the fairseq Transformer into Transformer-opt.

For each of our two language pair datasets, we first trained a baseline model from scratch. We then iterated over the five implemented techniques, training a model that implemented only that technique. Each of the 12 models (baseline plus five techniques per bitext) was trained for 150 passes over the sampled dataset. In the case of RASP and mBART, an additional pre-training phase of 150 passes over the pre-training dataset was run beforehand.

With all single-technique models trained and evaluated, to determine whether the positive impact of individual techniques could compound to produce an even higher BLEU score, we ran a final experiment combining multiple techniques with positive results to explore any further improvements over the baseline.

4.3 Evaluation

During training, the model was validated every 10 passes using its sacreBLEU score (Post, 2018) over the validation dataset, and we maintained the best checkpoint over the training period for final evaluation. This best checkpoint is then evaluated to produce a final sacreBLEU score, which we report in Table 2.

Models whose sacreBLEU score was higher than the baseline model were tested for statistical significance using bootstrap resampling ($n = 1000, p \leq 0.05$), using Compare-mt's significance testing utilities (Neubig et al., 2019). Compare-mt was also used to analyze the strengths of each technique, and we wrote additional testing tools as needed to analyze specific techniques.

5 Results

5.1 Isolated Results

The final sacreBLEU scores for each model are reported in Table 2, with statistically significant improvements highlighted in bold. Our implementations can be found on GitHub [1], along with scripts that can be used to reproduce all results described in this paper.

Model	De→En	Ne→En
Baseline	18.62	1.07
BART	**22.27**	**2.11**
JD	**20.09**	1.11
POS Tags	**19.64**	1.04
RASP	**19.26**	1.08
Data Diversification	11.42	0.09

Table 2: sacreBLEU scores of each technique used in isolation.

We note that, with all of our results, the translation quality in the Nepali-English translation task is significantly lower than that of the German-English translation task. This is due to a number of known factors, including the vast linguistic differences between Nepali and English (including a different alphabet, different word orders, and more), as well as the domain-specific nature of the FLORES dataset, which primarily consists of biblical text and technical documents including manuals from the Ubuntu OS (Guzmán et al., 2019).

5.1.1 RASP

RASP pre-training showed a statistically significant increase of 0.64 sacreBLEU in German-English translation, but provided no significant increase in

[1] https://github.com/gselzer/cs769-final-project

Nepali-English translation. Notably, in our implementation we pre-train only on noised source sentences, while (Lin et al., 2020) pre-train on noised and original source sentences. As shown in Table 3 we found no significant difference in translation quality when pre-training only on noised source sentences but instead discovered a boost to performance by halving the size of the pre-training data.

Model	**De→En**	**Ne→En**
RASP with original	19.27	0.84
RASP without original	19.26	1.08

Table 3: sacreBLEU scores for RASP models pre-trained with and without the inclusion of the original source sentences.

Of particular note in our tests was the ability of RASP to improve the sacreBLEU score of longer sentences, as shown in Figure 1. We hypothesize that this is due to the inclusion of more translated tokens in longer pre-training sentences, which strengthens the long-sequence capabilities of the model.

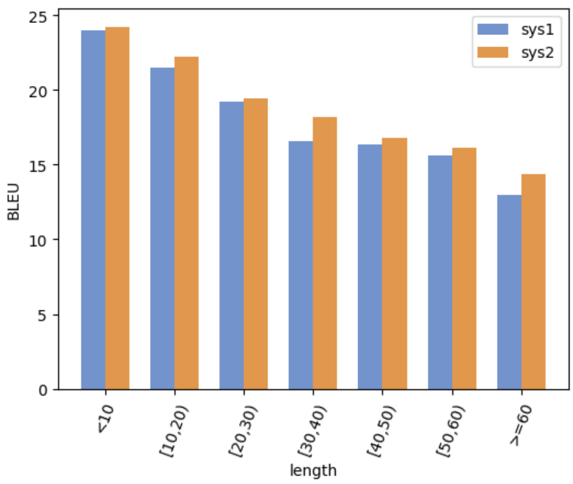

Figure 1: sacreBLEU vs. sentence length (in tokens); De→En with (orange) and without (blue) RASP pre-training

A likely reason behind the lack of improvement in the Nepali-English translation task lies in the poor quality of our Nepali-English dictionary. Using statistics taken during data noising in the data preprocessing state, we know that in the German-English translation task, RASP was able to replace approximately 30% of the source language tokens with their target language equivalent. For the Nepali-English translation task, RASP was only able to replace 5% of the words. We believe that fewer replacements resulted in RASP's minimal effect on Nepali-English translation, and thus conclude that for RASP to be effective, a strong bilingual dictionary is essential.

5.1.2 POS Tags

When added during training, interleaved POS tags in the source language data resulted in a statistically significant increase of 1.02 sacreBLEU on the German-English translation task but did not significantly change the sacreBLEU on the Nepali-English translation task.

Surprisingly, fine-grained analysis shows no analyzable difference between the effects of POS tags and RASP on our baseline model. To an extent, this makes some sense as both techniques add additional context about the meaning of affected words; RASP places word-level translations within the same context to encode additional information while POS tags provide additional information by adding new tag tokens, and it seems that our model utilizes both types of context to the same effect. Most importantly, neither mRASP nor POS tags aided in the Nepali-English translation task. This could be due to either the subpar performance of our Nepali POS tagger or the insufficient number of sentences needed to utilize the information contained within the added tags.

5.1.3 Data Diversification

Data Diversification led to decreased translation quality when applied to our low-resource context of 10k German-English and 20k Nepali-English sentence pairs, as shown in Table 2. Although (Nguyen et al., 2019) applied data diversification to a low-resource setting, which yielded increased performance, their English-Nepali training set consisted of 500k sentence pairs, 25 times the size of our Nepali-English training set. Thus, the poor performance of data diversification in our tests is likely attributed to far less training data. Secondly, we only trained the synthetic data generation models for 60 epochs in order to keep model training time short, in accordance with our 3-hour training time limit. This may not have been enough epochs for the synthetic data generation model to converge, and therefore this technique is not suitable for our experimental settings. If the synthetic data generation models do not converge, then the synthetic translations being appended are of lesser quality than in the original dataset, negatively impacting future training.

5.1.4 Joint Dropout

JD led to a statistically significant increase in the test set sacreBLEU of 1.47 in the German-English translations, as shown in Table 2. This was in alignment with what the original authors of JD found (Araabi et al., 2023). However, it led to minimal gains on the Nepali-English dataset. This may be due to the model struggling to learn more about the complex relationships of linguistically dissimilar languages (Nepali and English) on such a small dataset, and thus, JD provided a negligible added benefit.

An example translation output below in Table 4 demonstrates how training a low-resource model with JD can improve translation quality by preventing the model from over-relying on specific words and helping it learn context. In this case, the baseline model learns the relation of "mountain" to its surrounding words "Each" and "has its own individual personality." JD prevents this by encouraging the model to learn the sentence composition, as it then properly translated the word "iceberg."

	English Translation	**BLEU**
Ref	Each iceberg has its own individual personality.	
BL	So every mountain has its individual personal personality.	23.9
BL+JD	Everybody iceberg has its own individual personality.	85.9

Table 4: A translation example where the De→En baseline (BL) model with JD outperformed the baseline model.

5.1.5 mBART

mBART pre-training led to statistically significant sacreBLEU gains on both German-English (+3.65) and Nepali-English (+1.04) translation. As seen in Figure 2, sacreBLEU gains over the baseline model increase proportionally with sentence length; this trend, shared with RASP, likely stems from the increased number of alterations performed as sentence length increases.

mBART showed the most success out of all five studied techniques and was the only technique able to improve Nepali-English translation as shown in Figure 3. We believe it outperformed RASP, POS tags, and JD as it does not rely on external language resources (bilingual dictionaries, POS Taggers, and alignment tools, respectively) which are of lesser quality for low-resource languages.

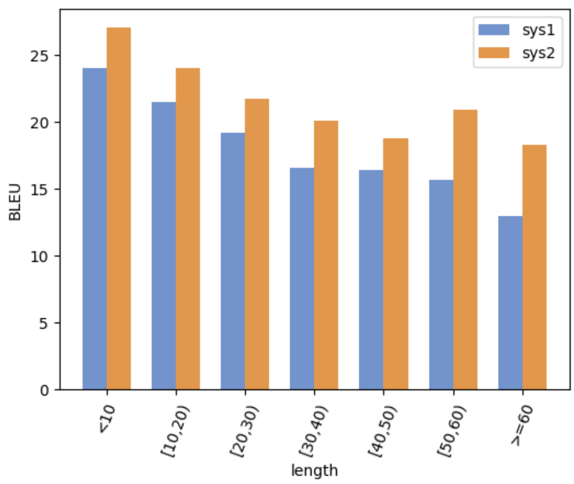

Figure 2: sacreBLEU vs. sentence length (in tokens); De→En with (orange) and without (blue) mBART pre-training

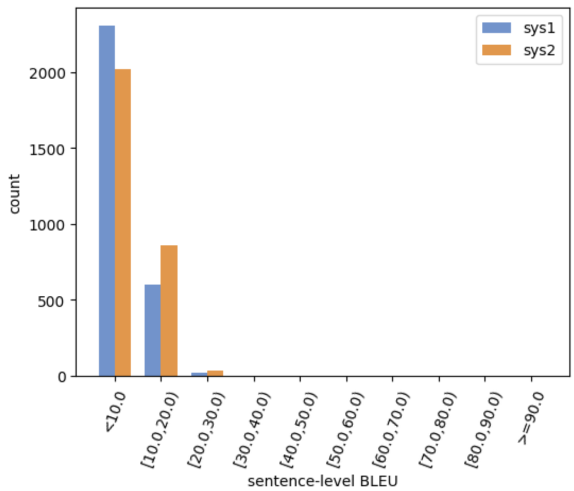

Figure 3: Count vs. sacreBLEU score range; Ne→En with (orange) and without (blue) mBART pre-training

5.2 Compositional Results

From the above analysis we found that the studied techniques improved over the baseline model in similar ways. Therefore, we trained a final model combining mBART and JD, the two techniques showing the most isolated improvement, to uncover any further benefit. The combination also showed statistically significant increases in sacreBLEU score compared to the baseline, as shown in Table 5. As shown in Figure 4, the combined model shows less decline in sacreBLEU as sentence length increases compared to either technique alone; the sacreBLEU score derived from the longest predicted translations is approximately only 4 less than the sacreBLEU derived from the shortest predicted translations. For the baseline model, the drop is nearly 10 sacreBLEU.

Model	De→En	Ne→En
Baseline	18.62	1.07
BART+JD	**21.92**	**2.13**

Table 5: sacreBLEU score for combined mBART and JD model

Also shown in Figure 4, the sacreBLEU score trend of the combined model bears a strong resemblance to that of mBART alone. We believe that the similarity in performance stems from a shared goal between these two techniques in utilizing context around masked tokens. Therefore, mBART and JD result in similar effects and add negligible benefit when combined. Notably, for translations with 60 or more tokens, the sacreBLEU score of the combined model was nearly 20, while for the mBART model, the sacreBLEU was around 18. This agrees with our hypothesis that these two techniques combine for similar effect and that the composition benefits longer sentences.

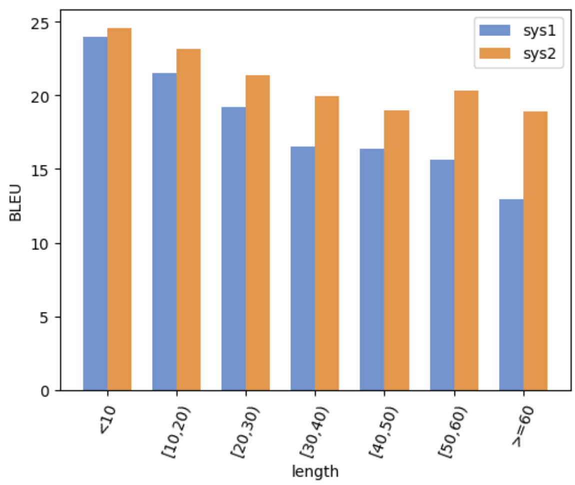

Figure 4: sacreBLEU vs. sentence length (in tokens); De→En with (orange) and without (blue) BART+JD pre-training

6 Conclusion

In this paper, we evaluate the proficiency of five different, modern techniques for low-resource MT on a small Transformer model. We have shown that mBART applies well to this scenario, producing statistically significant increases in the sacreBLEU score over baseline models. A large piece of this benefit likely comes from its independence from the source and target languages, which separates it from some of our other studied techniques. RASP, JD, Data Diversification and interleaving POS tags are unable to improve performance over the baseline in some or all scenarios. Most importantly, we have shown that all of these techniques can be used in an accessible and combined manner, making the widespread translation of low-resource languages, and iteration on existing models closer to reality.

The shortcomings of each technique could be the basis for its own future project. For example, we believe that RASP could be strengthened with the collection of robust, open bilingual dictionaries for low-resource languages; it would be worthwhile to determine whether a larger and more robust Nepali-English dictionary could replicate the gains we saw for our German-English translation task. Similarly, the compounding negative results seen in our Data Diversification experiments could be studied to determine the point at which the technique begins to negatively harm performance.

7 Contributions

Gabe transformed the code written in Homework 3 into the framework used in Homework 5 to enable reproducible training. He added RASP and POS tagging techniques to this new framework and executed many of the scripts for the group. Gabe also wrote code processing the Nepali-English bitext out of the FLORES dataset. Gabe wrote Python scripts used to call `compare-mt` on model outputs. In the report, Gabe wrote sections on RASP and POS Tagging, as well as the Experiments section, Abstract and Conclusion.

Luke implemented the Data Diversification and JD techniques, including the Python preprocessing and running files specific to Data Diversification. Luke executed the scripts for Data Diversification and JD and wrote the sections of the paper relating to the two techniques. Additionally, Luke wrote and ran the hypothesis test scripts to verify that our test results differed from the baseline model at a statistically significant level.

Zach focused primarily on the mBART technique. This work involved writing an mBART pipeline with a noising function and editing the preprocessing script to allow mBART as a step. Zach also modified the `compare-models` Python script to include significance testing, executed the significance testing, and wrote about mBART and its combination with JD in the report.

References

Antonios Anastasopoulos, Loïc Barrault, Luisa Bentivogli, Marcely Zanon Boito, Ondřej Bojar, Roldano Cattoni, Anna Currey, Georgiana Dinu, Kevin Duh, Maha Elbayad, Clara Emmanuel, Yannick Estève, Marcello Federico, Christian Federmann, Souhir Gahbiche, Hongyu Gong, Roman Grundkiewicz, Barry Haddow, Benjamin Hsu, Dávid Javorský, Věra Kloudová, Surafel Lakew, Xutai Ma, Prashant Mathur, Paul McNamee, Kenton Murray, Maria Nădejde, Satoshi Nakamura, Matteo Negri, Jan Niehues, Xing Niu, John Ortega, Juan Pino, Elizabeth Salesky, Jiatong Shi, Matthias Sperber, Sebastian Stüker, Katsuhito Sudoh, Marco Turchi, Yogesh Virkar, Alexander Waibel, Changhan Wang, and Shinji Watanabe. 2022. Findings of the IWSLT 2022 evaluation campaign. In *Proceedings of the 19th International Conference on Spoken Language Translation (IWSLT 2022)*, pages 98–157, Dublin, Ireland (in-person and online). Association for Computational Linguistics.

Stephen R Anderson. 2010. How many languages are there in the world?

Ali Araabi, Vlad Niculae, and Christof Monz. 2023. Joint dropout: Improving generalizability in low-resource neural machine translation through phrase pair variables.

Ondřej Bojar, Christian Buck, Christian Federmann, Barry Haddow, Philipp Koehn, Johannes Leveling, Christof Monz, Pavel Pecina, Matt Post, Herve Saint-Amand, Radu Soricut, Lucia Specia, and Aleš Tamchyna. 2014. Report on the 2014 wmt evaluation campaign. In *Findings of the 2014 Workshop on Statistical Machine Translation*, pages 12—-58, Baltimore, Maryland, USA.

Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter, Christopher Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. 2020. Language models are few-shot learners. In *Proceedings of the 34th International Conference on Neural Information Processing Systems*, NIPS'20, Red Hook, NY, USA. Curran Associates Inc.

Mauro Cettolo, Jan Niehues, Sebastian Stüker, Luisa Bentivogli, and Marcello Federico. 2014. Report on the 11th IWSLT evaluation campaign. In *Proceedings of the 11th International Workshop on Spoken Language Translation: Evaluation Campaign*, pages 2–17, Lake Tahoe, California.

Mauro Cettolo, Jan Niehues, Sebastian Stüker, Luisa Bentivogli, and Marcello Federico. 2013. Report on the 10th iwslt evaluation campaign. In *Proceedings of the 10th International Workshop on Spoken Language Translation: Evaluation Campaign*, Heidelberg, Germany.

Peng-Jen Chen, Ann Lee, Changhan Wang, Naman Goyal, Angela Fan, Mary Williamson, and Jiatao Gu. 2020. Facebook AI's WMT20 news translation task submission. In *Proceedings of the Fifth Conference on Machine Translation*, pages 113–125, Online. Association for Computational Linguistics.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186, Minneapolis, Minnesota. Association for Computational Linguistics.

Shuoyang Ding, Adithya Renduchintala, and Kevin Duh. 2019. A call for prudent choice of subword merge operations in neural machine translation. In *Proceedings of Machine Translation Summit XVII: Research Track*, pages 204–213, Dublin, Ireland. European Association for Machine Translation.

Francisco Guzmán, Peng-Jen Chen, Myle Ott, Juan Pino, Guillaume Lample, Philipp Koehn, Vishrav Chaudhary, and Marc'Aurelio Ranzato. 2019. The FLORES evaluation datasets for low-resource machine translation: Nepali–English and Sinhala–English. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 6098–6111, Hong Kong, China. Association for Computational Linguistics.

Barry Haddow, Rachel Bawden, Antonio Valerio Miceli Barone, Jindřich Helcl, and Alexandra Birch. 2022. Survey of low-resource machine translation. *Computational Linguistics*, 48(3):673–732.

Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. 2015. Distilling the knowledge in a neural network.

Guillaume Lample, Alexis Conneau, Marc'Aurelio Ranzato, Ludovic Denoyer, and Hervé Jégou. 2018. Word translation without parallel data. In *International Conference on Learning Representations*.

Mike Lewis, Yinhan Liu, Naman Goyal, Marjan Ghazvininejad, Abdelrahman Mohamed, Omer Levy, Ves Stoyanov, and Luke Zettlemoyer. 2019. Bart: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension.

Shaojun Li, Yuanchang Luo, Daimeng Wei, Zongyao Li, Hengchao Shang, Xiaoyu Chen, Zhanglin Wu, Jinlong Yang, Zhiqiang Rao, Zhengzhe Yu, Yuhao Xie, Lizhi Lei, Hao Yang, and Ying Qin. 2022. HW-TSC systems for WMT22 very low resource supervised MT task. In *Proceedings of the Seventh Conference on Machine Translation (WMT)*, pages 1098–1103, Abu Dhabi, United Arab Emirates (Hybrid). Association for Computational Linguistics.

Zehui Lin, Xiao Pan, Mingxuan Wang, Xipeng Qiu, Jiangtao Feng, Hao Zhou, and Lei Li. 2020. Pre-training multilingual neural machine translation by leveraging alignment information. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 2649–2663, Online. Association for Computational Linguistics.

Yinhan Liu, Jiatao Gu, Naman Goyal, Xian Li, Sergey Edunov, Marjan Ghazvininejad, Mike Lewis, and Luke Zettlemoyer. 2020. Multilingual denoising pre-training for neural machine translation.

Alireza Mohammadshahi, Vassilina Nikoulina, Alexandre Berard, Caroline Brun, James Henderson, and Laurent Besacier. 2022. Small-100: Introducing shallow multilingual machine translation model for low-resource languages.

Rudra Murthy, Anoop Kunchukuttan, and Pushpak Bhattacharyya. 2019. Addressing word-order divergence in multilingual neural machine translation for extremely low resource languages. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 3868–3873, Minneapolis, Minnesota. Association for Computational Linguistics.

Graham Neubig, Zi-Yi Dou, Junjie Hu, Paul Michel, Danish Pruthi, and Xinyi Wang. 2019. compare-mt: A tool for holistic comparison of language generation systems. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics (Demonstrations)*, pages 35–41, Minneapolis, Minnesota. Association for Computational Linguistics.

Xuan-Phi Nguyen, Shafiq R. Joty, Wu Kui, and Ai Ti Aw. 2019. Data diversification: An elegant strategy for neural machine translation. *CoRR*, abs/1911.01986.

Bista Nirooj, Nirmal Nepal, and Amit Shrestha. 2020. Nepdict. `https://github.com/nirooj56/Nepdict`.

Myle Ott, Sergey Edunov, Alexei Baevski, Angela Fan, Sam Gross, Nathan Ng, David Grangier, and Michael Auli. 2019. fairseq: A fast, extensible toolkit for sequence modeling. In *Proceedings of NAACL-HLT 2019: Demonstrations*.

F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay. 2011. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830.

Matt Post. 2018. A call for clarity in reporting BLEU scores. In *Proceedings of the Third Conference on Machine Translation: Research Papers*, pages 186–191, Brussels, Belgium. Association for Computational Linguistics.

Peng Qi, Yuhao Zhang, Yuhui Zhang, Jason Bolton, and Christopher D. Manning. 2020. Stanza: A python natural language processing toolkit for many human languages. *CoRR*, abs/2003.07082.

Víctor M. Sánchez-Cartagena, Juan Antonio Pérez-Ortiz, and Felipe Sánchez-Martínez. 2020. Understanding the effects of word-level linguistic annotations in under-resourced neural machine translation. In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 3938–3950, Barcelona, Spain (Online). International Committee on Computational Linguistics.

Liling Tan. 2014. Phrase extraction algorithm.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. *CoRR*, abs/1706.03762.

Marion Weller-di Marco and Alexander Fraser. 2022. Findings of the WMT 2022 shared tasks in unsupervised MT and very low resource supervised MT. In *Proceedings of the Seventh Conference on Machine Translation (WMT)*, pages 801–805, Abu Dhabi, United Arab Emirates (Hybrid). Association for Computational Linguistics.

Robert Östling and Jörg Tiedemann. 2016. Efficient word alignment with markov chain monte carlo. *The Prague Bulletin of Mathematical Linguistics*, 106(1):125–146.